

Industrial Enclosure Cooling — A Technical White Paper

HCK Cabinet Air Conditioners · Zhejiang Haocheng Industrial Control Equipment Co., Ltd. · Version 2026 · CE-marked · OEM/ODM

Abstract. Sealed electrical and control enclosures must reject internal heat while excluding dust, moisture and corrosive atmospheres. This paper explains closed-loop cabinet air conditioning, a reproducible cooling-load method, the HCK product range (2000–7500 W), and where these units are applied — from energy-storage cabins and prefabricated substations to base stations, EV-charging cabinets, mining sites and offshore platforms.

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1. Why sealed enclosure cooling matters

Power electronics, drives, transformers and batteries inside a cabinet convert part of their input energy into heat. Without active thermal management, internal temperature rises until components derate, age prematurely or trip. Forcing ambient air with a fan solves heat but admits dust and humidity; a *closed-loop cabinet air conditioner* cools the internal air while the enclosure stays sealed at IP55, protecting sensitive equipment in harsh environments.

2. How a closed-loop cabinet air conditioner works

A cabinet AC is a self-contained refrigeration loop mounted on the enclosure wall. Internal air passes over the cold evaporator; heat is pumped to the external condenser and rejected outdoors. Key elements:

- **Rotary compressor** — compact, low-vibration, suitable for continuous duty.
- **R410A refrigerant** — zero-ozone-depletion, high efficiency.
- **High-efficiency motor** and electronic control with **RS485 / Modbus** for remote monitoring.
- **Wide operating range –40...+55 °C** for outdoor and cold-climate sites.
- **No US-restricted components** — supply-chain safe for sanctioned markets.

3. Cooling-load calculation (reproducible method)

Total cooling demand is the sum of internal heat and envelope heat gain:

$$Q_{\text{total}} = Q_{\text{internal}} + Q_{\text{envelope}}$$

- **Q_{internal} (W)** — nameplate heat dissipation of all installed equipment (drives, PLC, rectifiers, batteries).

- **Q_{envelope} (W) = U × A × ΔT** — where *U* is the enclosure's thermal transmittance (≈6.0 W/m²K bare steel, ≈3.0 light insulation, ≈1.8 well insulated), *A* its surface area, and Δ*T* = (max ambient – target internal temperature).

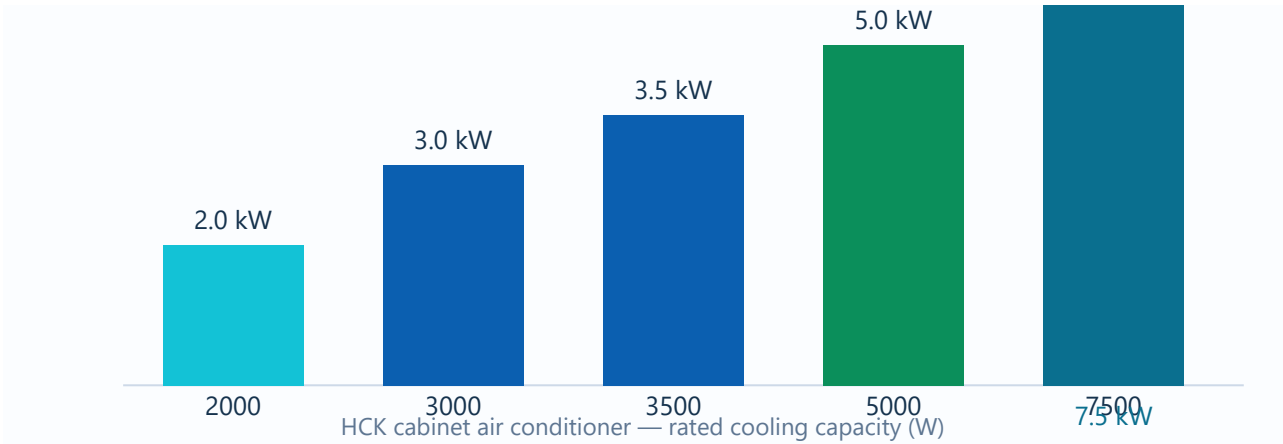
Worked example. A 0.6 m³ cabinet (≈800×600×1200 mm, A≈5.5·V^{2/3}≈4.3 m²), internal load 450 W, ambient 50 °C, target 35 °C, bare steel (U=6.0): Q_{envelope} = 6.0 × 4.3 × 15 ≈ 387 W; Q_{total} ≈ 837 W; with a 1.3 safety factor the required capacity is ≈1088 W — a small HCK unit covers it. For larger drives or hotter sites, step up to 3000–7500 W.

A free interactive version of this method is available: [HCK Cabinet Air Conditioner Selector](#).

4. HCK product range & specifications

Model	Cooling capacity	Supply	Protection	Operating range	Refrigerant	Comms
HCK-2000*	≈2000 W	220 V ±10% 50 Hz / 48 V DC opt.	IP55	–40...+55 °C	R410A	RS485
HCK-3000	≈3000 W	220 V ±10% 50 Hz / 48 V DC opt.	IP55	–40...+55 °C	R410A	RS485
HCK-3500	≈3500 W	220 V ±10% 50 Hz / 48 V DC opt.	IP55	–40...+55 °C	R410A	RS485
HCK-5000	≈5000 W	220 V ±10% 50 Hz / 48 V DC opt.	IP55	–40...+55 °C	R410A	RS485
HCK-7500	≈7500 W	220 V ±10% 50 Hz / 48 V DC opt.	IP55	–40...+55 °C	R410A	RS485

*HCK-2000 is in final design/qualification. All models: rotary compressor, high-efficiency motor, corrosion-protected coil, CE-marked, available as OEM/ODM.



5. Selection guide (step by step)

1. List internal heat loads (W) of all devices → **Q_{internal}**.
2. Measure or estimate enclosure surface area *A* and insulation → *U*.
3. Set max ambient and target internal temp → ΔT ; compute **Q_{envelope}**.
4. Add them, apply a 1.2–1.3 safety factor → **required capacity**.
5. Pick the smallest HCK model with rated capacity $\geq$ required. Above 7500 W, request a custom solution.

6. Application scenarios

Energy-storage cabin (储能舱)

Prefabricated substation / cabin (预制舱)

Telecom base station (基站)

EV charging cabinet (充电桩)

Mining site (矿山)

Offshore / marine (海上)

All share the same need: keep sealed enclosures within component limits where dust, salt, humidity and temperature extremes would defeat passive cooling. HCK units are built for IP55, wide-temperature and corrosive atmospheres.

7. Design features

- **IP55 sealed loop** — dust- and jet-water-protected; internal air never mixes with ambient.
- **–40...+55 °C operation** — cold-start and desert-rated.
- **Corrosion protection** — coated coil and hardware for coastal/mining duty.
- **RS485 / Modbus** — alarm, temperature and run-state telemetry to SCADA/BMS.
- **Sanction-safe BOM** — rotary compressor, R410A, high-efficiency motor; no US-restricted components.

8. Cabinet AC vs. alternatives

Criterion	Cabinet AC (HCK)	Air-to-air heat exchanger	Filtered fan
Sealing / IP	IP55 closed loop	IP55 (no active cooling of hot days)	IP low (open)
Cooling below ambient	Yes (refrigeration)	No (limited by ambient)	No
Dust / humidity exclusion	Excellent	Good	Poor
Best use	Harsh / hot / sealed	Mild climate, low ΔT	Clean, cool indoor

9. FAQ

Q: Can HCK run on DC (solar / battery)?

A: Yes — 48 V DC supply option is available alongside 220 V $\pm 10\%$ 50 Hz.

Q: Is the unit certified to CE?

A: HCK cabinet air conditioners are CE-marked; certification documentation is provided. (Confirm latest status per model.)

Q: Can it be customized (OEM/ODM)?

A: Yes — voltage, dimensions, coating, communication and branding can be tailored.

Q: How is it monitored?

A: RS485 / Modbus exposes temperature, alarms and run state to plant BMS/SCADA.

Cite this document: Zhejiang Haocheng Industrial Control Equipment Co., Ltd. (2026). *Industrial Enclosure Cooling — A Technical White Paper*. HCK Cabinet Air Conditioners. <https://www.zjhcc.com/en/whitepaper.html>

10. Compliance & supply chain

HCK units use a **rotary compressor, R410A refrigerant and high-efficiency motor with no US-restricted components**, making them suitable for markets with sanctions constraints, including Iran (settled via CNY through Kunlun Bank). Products are CE-marked and offered as OEM/ODM.

11. About HCK

Zhejiang Haocheng Industrial Control Equipment Co., Ltd. (brand **HCK**) designs and manufactures cabinet, box-type and prefabricated-cabin air conditioners from 2000 to 7500 W. Visit www.zjhcc.com or contact the engineering team for selection support.